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ON A CONJECTURE OF GEORGE BECK. II

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Abstract. In this paper, we give a combinatorial proof of a generating function identity concerning the sum of the smallest parts in the distinct partitions of n.

1. Introduction

As usual, a *partition* of a positive integer n is a weakly decreasing sequence of positive integers whose sum equals n.

In my previous paper [5], I proved the following conjecture due to George Beck (cf. A034296 in [7]).

Theorem 1.1. *The number of gap-free partitions (i.e. partitions with the difference between each pair of consecutive parts being at most* 1*) of n is also the sum of the smallest parts in the distinct partitions (i.e. partitions with distinct parts) of n with an odd number of parts.*

Let $\mathscr{DO}$ denote the set of distinct partitions with an odd number of parts. The main idea in [5] is to use the differentiation technique to study the generating function of $\mathrm{sspt}_{\mathscr{DO}}(n)$, the sum of the smallest parts in the partitions of n in $\mathscr{DO}$. More precisely, I showed that

$$\sum\nolimits_{n\geq 1} \mathrm{sspt}_{\mathscr{DO}}(n)q^n = \sum\nolimits_{t\geq 1} (q^t/(1-q^t))(-q)_{t-1}, \tag{1.1}$$

where and in what follows, we use the standard q-series notation

$$(a)_n = (a;q)_n := \prod_{k=0}^{n-1}(1-aq^k) \qquad \text{and} \qquad (a)_\infty = (a;q)_\infty := \prod_{k\geq 0}(1-aq^k).$$

In a subsequent paper [8], Yang also provided a combinatorial proof of Beck's conjecture.

Let $\mathscr{D}$ be the set of distinct partitions. Here we will not include the empty partition unless otherwise specified.

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For any partition π, we denote by $|\pi|$ the sum of the parts of π, by $\Lambda(\pi)$ the largest part of π, by $\sigma(\pi)$ the smallest part of π, and by $\sharp(\pi)$ the number of parts of π.

The following general result is almost shown in [5] (in which the cases $z = \pm 1$ are proved).

Theorem 1.2. *It holds that*

$$\sum_{\pi\in\mathscr{D}} \sigma(\pi) z^{\sharp(\pi)} q^{|\pi|} = \sum_{t\geq 1} (q^t/(1-q^t))\big((-z)_t - 1\big). \tag{1.2}$$

The proof of this generating function identity can still be deduced by the differentiation technique. In fact, it is essentially the same as the proof of (2.3) in [5]:

$$(-q)_\infty \sum_{m\geq 1} (mq^m/(-q)_m) = \sum_{t\geq 1} (q^t/(1-q^t))\big((-1)_t - 1)\big)$$

together with the observation on p. 648 of [5]:

$$\sum_{\pi\in\mathscr{D}} \sigma(\pi) q^{|\pi|} = \sum_{m\geq 1} mq^m(-q^{m+1})_\infty = (-q)_\infty \sum_{m\geq 1} (mq^m/(-q)_m).$$

In this paper, we will focus on the combinatorial viewpoint.

2. A combinatorial approach

Our starting point is the following double counting argument, which appears to be able to be adapted to many types of partition sets.

For a nonnegative integer t, we define

$$\mathscr{D}_t := \big\{\pi \in \mathscr{D} \ : \ \Lambda(\pi) \geq t+1 \text{ and } \Lambda(\pi) - \sigma(\pi) \leq t\big\}.$$

Now given any $\pi \in \mathscr{D}$, if $\pi \in \mathscr{D}_t$, then $\Lambda(\pi) - \sigma(\pi) \leq t \leq \Lambda(\pi) - 1$ by the definition. Hence, π is exactly contained in the following $\sigma(\pi)$ partition sets: $\mathscr{D}_{\Lambda(\pi)-1}$, $\mathscr{D}_{\Lambda(\pi)-2}$, $\ldots$, $\mathscr{D}_{\Lambda(\pi)-\sigma(\pi)}$. We therefore have

Theorem 2.1. *It holds that*

$$\sum_{\pi\in\mathscr{D}} \sigma(\pi) z^{\sharp(\pi)} q^{|\pi|} = \sum_{t\geq 0} \sum_{\pi\in\mathscr{D}_t} z^{\sharp(\pi)} q^{|\pi|}. \tag{2.1}$$

One immediately sees that the remaining task is to study the generating function for $\mathscr{D}_t$ with $t \geq 0$.

We remark that, for certain partition sets, if we only require the difference between the largest and smallest parts to be bounded by t, then the generating functions are studied in a series of papers [1–4,6]. In particular, in my joint work with Yee [6], we provided a combinatorial approach that can be easily adapted to obtain the generating function for $\mathscr{D}_t$.

For convenience, we now consider the generating function for $\mathscr{D}_{t-1}$ with $t \geq 1$.

Let $\mathscr{B}_t$ be the set of partition pairs (μ, ν) where μ is nonempty and its parts all have size t, and ν is a nonempty distinct partition with 0 being allowed as a part and the largest part being at most $t-1$. For example,

$$\big((5,5,5,5,5),\ (4,2,1,0)\big) \in \mathscr{B}_5.$$

Furthermore, we use $|(\mu,\nu)|$ to denote $|\mu| + |\nu|$.

For $\pi = (\pi_1, \pi_2, \ldots, \pi_\ell)$ in $\mathscr{D}_{t-1}$, we put $s = \lfloor \pi_\ell / t \rfloor$ with $\lfloor x \rfloor$ being the conventional floor function. We also let k be the positive integer such that $\pi_k \geq (s+1)t$ and $\pi_{k+1} < (s+1)t$. If there is no such k, then we let $k = 0$.

Now we construct a map $\phi_t : \mathscr{D}_{t-1} \to \mathscr{B}_t$ defined by

$$\begin{aligned}\phi_t\colon &(\pi_1, \pi_2, \ldots, \pi_\ell)\\ &\mapsto \big((\underbrace{t, t, t, \ldots, t}_{\substack{s(\ell-k)+(s+1)k\\ \text{times}}}), (\pi_{k+1} - st, \ldots, \pi_\ell - st, \pi_1 - (s+1)t, \ldots, \pi_k - (s+1)t)\big).\end{aligned}$$

Note that the second subpartition can be treated as $(\pi_1, \pi_2, \ldots, \pi_\ell)$ reduced modulo t, cyclically permuted such that they are weakly decreasing.

Similar to Theorem 2.1 of [6], we have

Lemma 2.2. *ϕ_t is a weight preserving map from $\mathscr{D}_{t-1}$ to $\mathscr{B}_t$. Furthermore, the number of parts is preserved by the second subpartition of the image.*

Proof. Let $(\mu, \nu) = \phi_t(\pi)$. We first show that μ is nonempty. Since $\pi \in \mathscr{D}_{t-1}$, we have $\pi_1 \geq (t-1) + 1 = t$. Hence we take out at least one t from π_1 to form μ, which implies that μ is not empty.

On the other hand, we know that π is a distinct partition. Since $\pi_1 - \pi_\ell \leq t - 1 < t$, $s = \lfloor \pi_\ell / t \rfloor$, and $\pi_k \geq (s+1)t > \pi_{k+1}$, we have

$$t > \pi_{k+1} - st > \cdots > \pi_\ell - st > \pi_1 - (s+1)t > \cdots > \pi_k - (s+1)t.$$

Note that $\pi_k - (s+1)t$ could be 0 since π_k could be $(s+1)t$. Hence ν satisfies the conditions. It follows that $(\mu, \nu) \in \mathscr{B}_t$.

At last, it is obvious from the definition of ϕ_t that $|\phi_t(\pi)| = |\pi|$ and $\sharp(\nu) = \sharp(\pi)$. $\square$

The rest of the argument is different to that in [6]. We shall show

Lemma 2.3. *ϕ_t is invertible.*

Proof. Let $(\mu, \nu) \in \mathscr{B}_t$. Let the number of t in μ be $r \geq 1$ and let $\nu = (\nu_1, \nu_2, \ldots, \nu_\ell)$. Now we write $r = m\ell + r^*$ with $m \geq 0$ and $0 \leq r^* \leq \ell - 1$ being integers. We construct the inverse $\phi_t^{-1} : \mathscr{B}_t \to \mathscr{D}_{t-1}$ as follows.

$$\phi_t^{-1}\colon (\mu, \nu) \mapsto (\nu_{\ell-r^*+1} + (m+1)t, \ldots, \nu_\ell + (m+1)t, \nu_1 + mt, \ldots, \nu_{\ell-r^*} + mt).$$

We now show that the image is in $\mathscr{D}_{t-1}$. Recall that $0 \le \nu_\ell < \cdots < \nu_1 \le t-1$. If $r^* \neq 0$, since $\nu_\ell + t > \nu_1$, we have

$$\nu_{\ell-r^*+1} + (m+1)t > \cdots > \nu_\ell + (m+1)t > \nu_1 + mt > \cdots > \nu_{\ell-r^*} + mt.$$

Notice that $\nu_{\ell-r^*+1} + (m+1)t \ge t$. We further notice that $\nu_{\ell-r^*}$ is not the smallest part of ν, and hence $\nu_{\ell-r^*} > 0$. At last, we have $(\nu_{\ell-r^*+1} + (m+1)t) - (\nu_{\ell-r^*} + mt) = t - (\nu_{\ell-r^*} - \nu_{\ell-r^*+1}) \le t-1$. Hence in this case the image is in $\mathscr{D}_{t-1}$.

If $r^* = 0$, then $m \ge 1$ since $r \ge 1$. We have $\nu_1 + mt > \cdots > \nu_\ell + mt > 0$ and $\nu_1 + mt \ge t$. We also have $(\nu_1 + mt) - (\nu_\ell + mt) = \nu_1 - \nu_\ell \le t-1$. Hence the image is also in $\mathscr{D}_{t-1}$.

From the definition of ϕ_t and ϕ_t^{-1}, it is apparent that $\phi_t^{-1}(\phi_t(\pi)) = \pi$. Hence ϕ_t is invertible. $\square$

Example 2.1. For the partition sets $\mathscr{D}_4$ and $\mathscr{B}_5$, we have

$$(9,7,6,5) \underset{\phi_5^{-1}}{\overset{\phi_5}{\rightleftharpoons}} \big((5,5,5,5),\ (4,2,1,0)\big) \qquad \text{and}$$

$$(10,9,7,6) \underset{\phi_5^{-1}}{\overset{\phi_5}{\rightleftharpoons}} \big((5,5,5,5,5),\ (4,2,1,0)\big).$$

It follows from Lemmas 2.2 and 2.3 that ϕ_t is a bijection from $\mathscr{D}_{t-1}$ to $\mathscr{B}_t$. Hence, for $t \ge 1$,

$$\sum\nolimits_{\pi \in \mathscr{D}_{t-1}} z^{\sharp(\pi)} q^{|\pi|} = \sum\nolimits_{(\mu,\nu) \in \mathscr{B}_t} z^{\sharp(\nu)} q^{|\mu|+|\nu|}. \tag{2.2}$$

The generating function for $\mathscr{B}_t$ is easy to get:

$$\sum\nolimits_{(\mu,\nu) \in \mathscr{B}_t} z^{\sharp(\nu)} q^{|\mu|+|\nu|} = (q^t/(1-q^t))\big((-z)_t - 1\big), \tag{2.3}$$

where $q^t/(1-q^t)$ comes from the first subpartition whereas $(-z)_t - 1$ comes from the second subpartition. Consequently, we have

Theorem 2.4. *For $t \ge 1$, it holds that*

$$\sum\nolimits_{\pi \in \mathscr{D}_{t-1}} z^{\sharp(\pi)} q^{|\pi|} = (q^t/(1-q^t))\big((-z)_t - 1\big). \tag{2.4}$$

Together with (2.1), we have

$$\sum\nolimits_{\pi \in \mathscr{D}} \sigma(\pi) z^{\sharp(\pi)} q^{|\pi|} = \sum\nolimits_{t \ge 1} (q^t/(1-q^t))\big((-z)_t - 1\big),$$

which completes the proof of Theorem 1.2.

3. Closing remarks

As I showed in [5], (1.1) follows since

$$\sum\nolimits_{n\geq 1} \mathrm{sspt}_{\mathscr{DO}}(n)q^n = \frac{1}{2}\sum\nolimits_{\pi\in\mathscr{D}} \sigma(\pi)\Big(1^{\sharp(\pi)} - (-1)^{\sharp(\pi)}\Big)q^{|\pi|}.$$

Note that the $z = 1$ and $z = -1$ cases of (1.2) respectively correspond to (2.3) and (2.2) in [5].

I was also pointed out by Dazhao Tang that another conjecture of Beck, which is proposed in [7, A237665], can be proved in the same way.

Conjecture 3.1 (Beck)**.** *The number of gap-free partitions of n with at least two different parts is also the sum of the smallest parts in the distinct partitions of n with an even number of parts.*

Theorem 3.2. *Conjecture 3.1 is true.*

Proof. Let $\mathrm{sspt}_{\mathscr{DE}}(n)$ denote the sum of the smallest parts in the distinct partitions of n with an even number of parts. We have

$$\begin{aligned}\sum_{n\geq 1} \mathrm{sspt}_{\mathscr{DE}}(n)q^n &= \frac{1}{2}\sum_{\pi\in\mathscr{D}} \sigma(\pi)\Big(1^{\sharp(\pi)} + (-1)^{\sharp(\pi)}\Big)q^{|\pi|}\\ &= \sum_{t\geq 1}(q^t/(1-q^t))\big((-q)_{t-1} - 1\big).\end{aligned}$$

We next observe that the conjugates of gap-free partitions with at least two different parts are partitions with at least two different parts where only the largest part may repeat.

Let $\mathrm{gf}_2(n)$ count the number of gap-free partitions of n with at least two different parts. We have

$$\begin{aligned}\sum_{n\geq 1} \mathrm{gf}_2(n)q^n &= \sum_{t\geq 1}(q^t/(1-q^t))(-q)_{t-1} - \sum_{t\geq 1}(q^t/(1-q^t))\\ &= \sum_{t\geq 1}(q^t/(1-q^t))\big((-q)_{t-1} - 1\big).\end{aligned}$$

Here $\sum_{t\geq 1}(q^t/(1-q^t))(-q)_{t-1}$ is the generating function of partitions where only the largest part may repeat, and $\sum_{t\geq 1}(q^t/(1-q^t))$ is the generating function of partitions with only one different part. We conclude that $\mathrm{gf}_2(n) = \mathrm{sspt}_{\mathscr{DE}}(n)$. □

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